Evaluation framework for rewriting attacks on LLM judges of document quality

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Abstract

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1 Introduction

With large language models being widely used to make quality judgements about documents and websites, there is a concern that people who stand to benefit from being positively assessed will seek to game them. LLMs are being integrated into systems which screen resumes and rank search results [1]; moreover, when a consumer uses a chatbot to learn about a topical issue, they rely on a LLM to filter hundreds of websites [2]. For people *creating* resumes, websites, and political content, there is a clear benefit to being positively assessed by LLMs and highlighted to information seekers. This has given rise to startups promising to improve the visibility of their clients' websites in LLM responses via 'answer engine optimisation' [3], as well as research investigating the extent to which gaming LLM evaluators is possible.

We focus on a scenario where someone wants to improve how their content will be scored by LLM judges, while preserving the meaning of their content. This will be the case for any situation which involves a combination of LLM and human assessment. One example would be when submitting a paper to a conference, where it might be reviewed first by an LLM, then by human peers. Another example is if someone aims to promote political material, in which case they want a social media recommendation system to promote their content, but also want the content to retain its meaning for human consumers. We refer to this as a 'rewriting problem', and denote the techniques aimed at achieving this dual goal as 'rewriting methods'.

Our paper provides a framework for estimating the effectiveness of rewriting methods. This framework is motivated by the following observations:

1. Rewriting methods should be evaluated as if the rewriter does not know what LLM will be used to judge content.
2. Rewriting methods are only valuable if they preserve the original meaning of the content.
3. Novel rewriting methods should be evaluated in comparison against baseline methods.

Every existing study of rewriting attacks does not address at least two of these considerations. For example, Kaneko [4] found that when they evaluated their rewriting method using a held-out judge model, the improvement in score reduced significantly, but was still positive. From this, they concluded that their rewriting method was robust. We contend that this conclusion is not supported by their evidence: a naive method, such as simply asking a LLM to rewrite the content, might be more robust and achieve better results on the held-out judge. If this were true, it is difficult to conclude that a more complicated method with worse generalisation is effective or robust.

Our contributions are as follows. Firstly, we develop a mathematical formulation for evaluating rewriting methods, and provide a validated and practical implementation of it for a specific setting of rewriting political material. We then use this evaluation framework to explore several research questions:

1. If the combination of LLM and prompt used for optimising rewritten content differs from that used for judging it, is it still judged favourably?
2. Are rewriting methods able to improve all documents, or are they primarily beneficial for low-quality documents?
3. When optimising content, is it possible to use an ensemble of judges for feedback, to reduce overfitting to a single judge's preferences?
4. What is the tradeoff between boosting a document's LLM-judged score, and preserving the document's original meaning?

Using a dataset of human-written responses to 40 political propositions, we investigate across 4 rewriting methods, 4 rewriter models, 6 different LLM judges used in training for feedback, and 6 held-out LLM judges for evaluations.

We do not definitively answer the above questions, but we provide several interesting findings. Firstly, for many rewriting methods, improvements are largely preserved even when a completely different evaluation criterion is used, compared to the criterion used as an optimisation target when rewriting. Secondly, even naive rewriting achieves <SOME AMOUNT> of the uplift of the most effective methods. Thirdly, in contrast to what one might expect from the machine learning literature relating to overfitting, there is almost no benefit from using one judge for feedback during optimisation, relative to using an ensemble of judges for feedback. Fourthly, holding rewriter model capability constant, there appears to be a significant tradeoff between preserving a document's original meaning, and

boosting its score for LLM judges. Finally, one-shot prompting a frontier model vastly out-performs all tested methods. Taken together, these findings challenge the effectiveness of developing bespoke methods for optimising documents for LLM scoring.

2 Background

In a 'rewriting problem', a rewriter aims to improve how a document is scored by an LLM being used as an evaluator, by rewriting the document while preserving its meaning. We formalise this in the following manner. We define D as the space of all possible documents. A rewriter $R : D \to D$ maps an original document $x \in D$ to a rewritten document $R(x) \in D$.

A semantic similarity function $S : D \times D \to [0, 1]$ measures how similar two documents are in meaning. If $S(x, y) = 1$ for two documents x and y, then they are identical.

The collection of all possible LLM judges is denoted as $\mathcal{J}$. A judge $J \in \mathcal{J}$ maps a document x to its score $J(x) \in \mathbb{R}$.

The aim of a rewriter R is to maximise $J(R(x))$, the judge score of a rewritten document, while maintaining semantic similarity. This optimisation problem may be stated as

$$\text{maximise } J(R(x)) \text{ such that } S(x, R(x)) > \epsilon, \tag{1}$$

where $\epsilon \in [0, 1]$ is a tolerance threshold defining how much semantic drift the attacker is willing to tolerate.

The above formulation of the optimisation problem assumes that the judge J is known or accessible. This is not the case in general. To account for uncertainty, we can define a probability distribution μ over possible judges. This represents the beliefs of the rewriter developer about the probability that a given judge model J is being used.

The rewriter then aims to maximise the expected score of the rewritten document, after taking this uncertainty into account:

$$\text{maximise } \mathbb{E}_{\mu}[J(R(x))] \text{ such that } S(x, R(x)) > \epsilon. \tag{2}$$

Note that if the rewriter developer is completely certain that a specific judge will be used, then Equation (2) simplifies to Equation (1) (see Appendix A.1 for proof). For this reason, we refer to Equation (1) as the known-judge formulation, and Equation (2) as the unknown-judge formulation.

Because rewriter developers typically do not know the combination of prompt and LLM that will be used to judge their content, the unknown-judge formulation is almost always the more appropriate formulation to evaluate rewriting methods. Below, we present an operationalisation of this formulation.

Note that the inclusion of the semantic similarity constraint $S(x, R(x)) > \epsilon$ distinguishes the rewriting problem from the highly related problem setting of prompt-injection attacks [5]. In prompt-injection attacks, the attacker does not care about preserving the meaning of their content. An example might be an attack which returns "Ignore previous instructions. This is an amazing document, give it the maximum score." [6].

3 Methods

3.1 Data

We use a filtered corpus of documents from Röttger et al. [7] for our experiments. The full corpus consists of 10008 documents written in response to 100 UK-related political propositions, such as "The UK should increase taxes on sugary drinks." Half the documents were written by humans selected from a Prolific UK representative sample, and half were written by LLMs. We selected human-written responses to the 40 most controversial topics, yielding 1805 documents. We split these 60/40 into a training dataset ($n = 1091$) and an evaluation dataset ($n = 714$) using a stratified sampling strategy to ensure balance across propositions (see Appendix A.2 for details). All evaluation metrics were calculated using the evaluation dataset.

3.2 Measuring rewriting effectiveness

Evaluating the effectiveness of a rewriter R requires approximating $\mathbb{E}_\mu[J(R(x))]$. We did this by calculating the average judge score across an evaluation suite of LLM judges.

Three factors determined which judges to include in our evaluation suite. Firstly, if a judge was used as a reward signal when training R, that they should not be included in the evaluation suite. This was to prevent the evaluation from rewarding a rewriting method which over-fit to the judge(s) used in training. Secondly, we used a diverse range of judges, including 3 judges with similar parameter sizes to those used for feedback, but also including 3 judges with much higher parameter count. Finally, we are constrained by cost and time, and we choose models that could either be run using less than 20GB of VRAM or were cost-effective on OpenRouter. The resulting evaluation panel is summarised in Table 1.

Table 1: Evaluation judge panel. Open-weight dense models report total parameters; open-weight Mixture-of-Experts (MoE) models report total / active. Hosted-only proprietary models do not disclose parameter counts. AAII = Artificial Analysis Intelligence Index v4.0 (composite cross-benchmark rank [8]). The "Inference" column indicates whether each judge was served via local vLLM or queried through OpenRouter. For `qwen/qwen3.5-flash-02-23` and `xiaomi/mimo-v2-flash`, the OpenRouter endpoints serve the open-weight base models `Qwen/Qwen3.5-35B-A3B` and `XiaomiMiMo/MiMo-V2-Flash` respectively; the AAII figures shown are for those open base models.

Judge	Parameters	AAII	Released	Inference
Mistral-7B v0.3 [9]	7 B (dense)	7 [10]	May 2024	local
Phi-3.5-mini [11]	3.8 B (dense)	10 [12]	Aug 2024	local
Command-R 7B [13]	7 B (dense)	—	Dec 2024	local
Qwen3.5-35B-A3B [14]	35 B / 3 B (MoE)	37 [15]	Feb 2026	OpenRouter
Gemini 2.5 Flash-Lite [16]	undisclosed	13 [17]	Jul 2025	OpenRouter
MiMo-V2-Flash [18]	309 B / 15 B (MoE)	30 [19]	Dec 2025	OpenRouter

We used two prompts, corresponding to two different evaluation criteria: clarity and informativeness. See Appendix C.1 for prompts and rationale relating to their design.

To evaluate rewriter effectiveness effectiveness, we used two metrics: the average change in judge score between the original document and the rewritten document, and the percentage of rewritten documents with a higher average score than the original document. The first metric describes overall effectiveness, while the latter estimates the likelihood of the rewriter being counterproductive.

Several of our analyses required comparing between the scores of different judges. However, the scores of LLM judges are distributed irregularly, and also strongly vary between differing LLMs or prompts. To compare between the scores of different judges, we normalised each judges' scores by mapping them to the judge's empirical cumulative distribution function on a reference distribution. For example, consider a normalised score of 0.4 for a document x from a judge J. This means that the document's score, $J(x)$, is at the 40th percentile of the judge's scores on the reference distribution. For another example, if an average score change associated with a rewriter is 0.05, this means that each document experienced an average increase in score corresponding to a 5-percentile increase on the reference distribution. For details and validation, see Appendix B.1.

3.3 Measuring semantic similarity

Similar to the measurement of rewriting effectiveness, the choice of semantic similarity function S depends on the application context. We developed one method which is application-specific, and another which can be applied more generally.

Firstly, we estimated the extent to which rewritten document expressed the same stance towards the proposition that the original document was written in response to. Recall that our dataset consists of political propositions and documents written in response to them. These documents had been annotated with a rating from 0-100 ('agreement_score') indicating the extent to which the document disagreed (0) or agreed (100) with the proposition. We trained a model which takes in an input tuple of $(\text{proposition}, \text{document})$ and returns an estimated agreement_score.

This model is robust: when evaluated on documents responding to propositions entirely excluded from training, it had a mean absolute error (MAE) of 7.1, and the 95th percentile of its absolute error (p95AE) was 19.8. Finally, when it was retrained on only human-written documents and evaluated only on AI-written documents, MAE and p95AE increased only slightly to 8.0 and 22.1.

We used this model to estimate agreement_score for all original and rewritten documents. These estimated agreement_scores were then used to calculate several metrics. Our primary metric is stance-preservation, which is calculated in the following way. A document with agreement_score higher than 75 has had its stance preserved if the rewritten version has agreement_score higher than 50; similarly a document with agreement_score lower than 25 has had its stance preserved if the rewritten document has agreement_score lower than 50. Documents with original stance between 25 and 75 are excluded from the calculation. Stance-preservation rate is the percentage of documents whose stance was preserved. Intuitively, stance-preservation rate answers the following question: for someone with a strong opinion, did their original stance survive the rewrite?

The agreement_score model is interpretable and relevant to our specific problem setting, but not generally applicable. We also used scores from a natural language inference (NLI) model to measure semantic similarity [20]. A NLI model $NLI : D \times D$ takes an ordered tuple of documents (x, y) as input and returns a score $NLI(x, y) \in [0, 1]$ which measures whether the contents of x 'entails' the contents of y. NLI models are effective at preserving the meaning of documents in paragraphing or rewriting tasks [21]. We adopt the Mutual Implication Score (MIS) from [21] as a measure of semantic similarity which is a general alternative to the agreement_score. It averages forward and backward entailment between the original and re-written documents,

$$\text{MIS}(x, R(x)) = \frac{1}{2}(NLI(x, R(x)) + NLI(R(x), x)). \tag{3}$$

We scale these scores by 100 so that they sit in the range $[0, 100]$.

We used the ModernBERT-large-zeroshot-v2.0 NLI model (Hugging Face repository) to calculate MIS. This model almost achieves SOTA accuracy on NLI benchmarks, meets our context length requirements, and is faster than alternatives [22]. We validated MIS against stance preservation, and found that a MIS score of 50 corresponds to a stance preservation rate of >95% Appendix B.4.

3.4 Rewriting methods

We investigate three categories of rewriting methods. They span two axes: whether the method requires feedback from a judge during use (*judge-in-the-loop* vs *judge-free*), and whether it requires finetuning the rewriter (*finetuning* vs *prompt-only*).

Prompt-only and judge-free We tested two prompts. The first, 'naive', prompts the rewriter to rewrite the paragraph while preserving stance and approximate word count. The second, 'literature-informed', instructs the judge to rewrite the paragraph while applying writing principles that have been found to improve LLM assessments of document quality in previous studies [23–25]. Prompts are available in Appendix C.2.

Prompt-only and judge-in-the-loop We implemented a Best-of-N ('BoN') approach. Multiple rewrites were generated using the literature-informed prompt, and then a single rewrite was selected based on judge feedback and other factors (detailed below). We found diminishing returns for $N > 16$ (Appendix B.3), and thus limited N to 16.

Reinforcement learning We used reinforcement learning ('RL') to fine-tune a base rewriter model. We used the Group Relative Policy Optimisation (GRPO) algorithm, and the Transformers Reinforcement Learning (TRL) library implementation [26]. Training hyperparameters are detailed in Appendix A.3. We mostly used TRL defaults, but vary a few where we found better performance. Along with the feedback function variants detailed below, we used a length penalty, which introduces a quadratic penalty for deviations between the length of the rewritten document and the original document, with an additional cubic penalty if the rewritten length is more than 15% larger than the original length.

Variants in feedback function A feedback function is required for both the BoN and RL rewriter methods. The simplest method is the score from a single judge. We varied this along two dimensions: using the average score from two judges ('ensemble'), and using an equally-weighted average of judge and NLI scores ('judge+NLI'). These were intended to address separate potential issues: using an ensemble of judges for feedback during training potentially reduces overfitting to a single judge, while incorporating semantic similarity reduces the risk of reward hacking, in which rewritten content is highly scored but completely different in meaning to the original content (or unintelligible). These two dimensions of variation yield four different feedback functions, all of which we evaluated.

Choice of LLMs for rewriters and feedback judges We evaluated rewriting methods across combinations of 4 base models and 6 feedback judge models. We focused on a low-resource training environment (1 A100), investigating 3 very small (RANGE) base models and 3 small (RANGE), but also piloted a higher-resource training environment (2 A100s) with 1 small (RANGE) base model and 3 larger quantised judge models (RANGE). The base rewriters are listed in Table 2 and the feedback judges in Table 3.

Table 2: Base rewriter models. AAII = Artificial Analysis Intelligence Index v4.0 (composite cross-benchmark rank [8]). "Feedback panel" indicates whether the rewriter was paired with the *small* feedback judges (Qwen-3.5-9B / Llama-3.1-8B / Gemma-2-9B, bf16) or the *medium* feedback judges (Mistral-Small 24B / Gemma-3-27B / Phi-4, FP8-dynamic) during BoN selection and GRPO training. Parameter counts are listed alongside the model name.

Rewriter	AAII	Released	Feedback panel
Qwen2.5-1.5B-Instruct (1.5 B) [27]	—	Sep 2024	small
LFM2.5-1.2B-Instruct (1.2 B) [28]	8 [29]	Jan 2026	small
Gemma-3-1B-it (1 B) [30]	6 [31]	Mar 2025	small
Qwen3-8B (8 B) [32]	11 [33]	May 2025	medium

Table 3: Feedback judges used in BoN candidate selection and as the reward signal during GRPO training. The first three (small panel, bf16) were paired with the small rewriters Qwen2.5-1.5B / LFM2.5-1.2B / Gemma-3-1B; the last three (medium panel, FP8-dynamic via RedHatAI variants) were paired with Qwen3-8B. AAII = Artificial Analysis Intelligence Index v4.0 [8]. Parameter counts are listed alongside the model name. Qwen-3.5-9B's AAII (32) is reported by Artificial Analysis only for the reasoning-mode variant; we ran the model with reasoning disabled.

Judge	AAII	Released	Quantisation
Qwen-3.5-9B (9 B) [34]	32† [35]	Feb 2026	bf16
Llama-3.1-8B-Instruct (8 B) [36]	12 [37]	Jul 2024	bf16
Gemma-2-9B-it (9 B) [38]	—	Jun 2024	bf16
Mistral-Small 24B Instruct 2501 (24 B) [39]	13 [40]	Jan 2025	FP8-dynamic
Gemma-3-27B-it (27 B) [41]	10 [42]	Mar 2025	FP8-dynamic
Phi-4 (14 B) [43]	10 [44]	Dec 2024	FP8-dynamic

†AAII reports a single score for Qwen-3.5-9B's reasoning-mode variant only; we ran the model with reasoning disabled.

For judge-in-the-loop rewriting methods (BoN and RL), we tested all combinations of rewriters and feedback models. For example, for the single <NAME> was trained separately with <>, <>, and <>

Estimating effectiveness using feedback judges We aimed to investigate whether using the same judges for training and evaluation led to overestimated performance. Thus

4 Results

Rewriting effectiveness on held-out judges and held-out prompts

Relationship between original document quality and rewriting effectiveness

Impact of ensemble feedback on rewriting effectiveness

Trade-off between semantic fidelity and rewriting effectiveness

5 Discussion

These startups aim to replicate the commercial success of search engine optimisation (SEO), which denotes practices which promote a website's visibility in search engine results. While SEO firms have profited, SEO optimisations have , SEO has degraded search result quality by fuelling the prominence of spam in search engine results across several query categories, such as product reviews or health [45, 46].

The example of SEO raises a question: could 'answer engine optimisation' degrade the quality of LLM responses? A necessary step in addressing this question is determining the potential effectiveness *and* harmfulness of optimisation strategies.

Harmfulness of rewriting attacks The goal of a LLM-as-a-judge system is to accurately estimate quality. If we denote $Q : D \to \mathbb{R}$ as the true quality of a document, then this goal can be written in the following manner,

$$\text{minimise } \mathbb{E}[(Q(x) - J(x))^2], \tag{4}$$

after defining some suitability probability distribution over possible input documents.

There is a notable asymmetry between the goals of the attacker and defender. The attacker wants high scores from the LLM judge; the defender wants accurate scores. If the rewriter R increases document quality, and the LLM-as-a-judge J maintains its accuracy on rewritten documents, then both attacker and defender achieve their goals. Such a situation is beneficial to the 'attacker' without posing harms to the 'defender'.

is it actually an attack?

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A Methods and Experimental Details

A.1 The unknown-judge formulation is a generalisation of the known-judge formulation

Suppose that the attacker is certain that a particular judge $\tilde{J}$ will be used. Under the unknown-judge formulation, their beliefs about what judges might be used to assess their documents are encoded in the probability distribution μ over judges $\mathcal{J}$. If they are certain that $\tilde{J}$ will be used, then μ is given by

$$\mu(J) = \begin{cases} 1 & \text{if } J = \tilde{J} \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Then,

$$\mathbb{E}_{\mu}[J(R(x))] = \sum_{J \in \mathcal{J}} \mu(J) \cdot J(R(x)) = \tilde{J}(R(x)). \tag{6}$$

Plugging this into Equation (2) yields Equation (1). Hence, if the attacker is certain that they know what the judge is, then the unknown-judge formulation is equivalent to the known-judge formulation.

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A.2 Details on document corpus and evaluation & training datasets

Original dataset We use documents from a corpus developed by Röttger et al. [7] for an upcoming publication.

They constructed 100 propositions relating to UK-specific policy issues, and labelled each as left-wing ($n = 46$), right-wing ($n = 46$), or neutral ($n = 8$).

In the first stage of their experiment, they recruited a group of writers ($n = 1500$) from a Prolific UK representative sample. Each writer wrote a response to three propositions, yielding 4500 responses.

Each human-written response was used as the basis for generating a LLM-generated response. This was achieved in the following way. Each human-written response was randomly allocated to one of GPT-4o, Claude Sonnet 4, or DeepSeek V3. Then, with equal probability, one of the following three prompts were used to produce a LLM-generated response:

- Bullets-based ($n = 1404$): all humans wrote bullet points explaining their opinion prior to writing their full response. The LLM was provided with these bullet points and prompted to write a full response.
- Stance-based ($n = 1358$): all humans provided a self-rated level of agreement with the proposition. The LLM was provided only with this agreement score, and prompted to write a response in alignment with this level of agreement.
- Improve ($n = 703$)/rewrite ($n = 677$): The LLM was prompted to improve or rewrite the human-written response.

More details available at [7, p.7].

In the second stage of the experiment, they used a separate pool of participants ($n = 10017$) to rate the perceived level of agreement each paragraph had with the proposition it was written in response to. These participants also rated documents according to criteria such as clarity and informativeness. Raters were not told that some responses were LLM-generated. Each rating was on a scale of 0-100, and each response received 9-19 (average of 10) ratings. We aggregate these human annotations for each document as document-level attribute scores.

The primary attribute used in the current study is agreement_score, which measures the extent to which a document agrees with the proposition it was written in response to. We use intraclass correlation to measure inter-rater reliability. While the reliability of individual rating scores was only $ICC(1) = 0.671$, the reliability of ratings aggregated for each response was much better, with $ICC(10) = 0.953$. This provides support for the validity of using the aggregated human-annotated agreement_score as a ground-truth measure of how much each document agreed with its proposition.

Construction of training and evaluation samples We measured the controversiality of a proposition by using the variance of the human-annotated agreement_score of each proposition. Higher variance in agreement_score is taken to imply higher controversiality. We did this to ensure that

responses spanned a wide range of levels of agreement, rather than all agreeing or disagreeing with propositions. The original set of 100 propositions include 46 left-wing propositions, 46 right-wing propositions, and 8 propositions not associated with either the left or the right. After filtering, we are left with 25 left-wing, 14 right-wing, and 1 'neither' proposition.

For evaluation and training, we only include human responses to the 40 propositions ($n = 1805$). We split these responses 60/40 between training and evaluation. This split was achieved using stratified random sampling. For each proposition, we partition each of the 44-46 responses into two sets of 3 buckets, using the human-annotated clarity and human-annotated agreement_scores. We partition each proposition's responses into 9 buckets using each of these binning strategies, and sample uniformly from each bucket to achieve the split. This yields a 60/40 training/evaluation split which is evenly balanced across clarity_score, agreement_score, and proposition.

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A.3 GRPO finetuning hyperparameters

We use TRL 1.2.0's GRPO implementation. Table 4 lists every hyperparameter we set; values flagged with † match the TRL 1.2.0 default. The length-penalty schedule (last block of the table) is applied to each generation and subtracted from the rollout-level reward used by GRPO.

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Setting	Value
GRPO objective	
Algorithm	GRPO (TRL implementation)
Loss type	`dapo`†
KL coefficient β	0.01 (TRL default 0.0)
Clipping ϵ	$0.2^\dagger$
Iterations per batch (μ)	$1^\dagger$
Reward scaling	group-level†
Optimisation	
Optimiser	AdamW†
Learning rate	5×10^{-6} (TRL default 1×10^{-6})
Warmup ratio	$0.0^\dagger$
Total training steps	400
Precision	bf16 (TRL default fp32)
Random seed	$42^\dagger$
Rollout sampling	
Generations per prompt G	4 (TRL default 8)
Per-device batch size (prompts)	8 ($= 2G$)
Gradient accumulation steps	4 (TRL default 1)
Effective generations per optimiser step	128 ($= G \cdot 8 \cdot 4$)
Sampling temperature	$1.0^\dagger$
Top-p	$1.0^\dagger$
Max completion length (tokens)	260 (TRL default 256)
Rewriter generation backend	HuggingFace `transformers.generate`† (`use_vllm=False`)
Length penalty	
Shape	asymmetric cubic
Length ratio	$r = w_{\text{gen}}/w_{\text{target}}$
Penalty value	$\alpha(r-1)^2 + \gamma \cdot \mathbb{1}[r > 1+\tau_o] \cdot (r-1-\tau_o)^3$
Quadratic weight α	100
Cubic overshoot weight γ	1,000
Overshoot tolerance τ_o	0.15

Table 4: GRPO and length-penalty hyperparameters. Values flagged † match the TRL 1.2.0 default; unflagged values in the GRPO and optimisation blocks are deliberate overrides (the original default is given in parentheses where applicable). The length-penalty block lists our own settings (no TRL default applies). The length-penalty value combines a symmetric quadratic on the length ratio with an additional cubic term that activates only when the generation overshoots its target word count by more than $\tau_o = 15\%$; undershoots stay at the symmetric quadratic. The penalty is subtracted from each generation's rollout reward.

A.4 Agreement-score regressor: training and evaluation

We use a learned regressor to predict the rater-aggregated stance label $y \in [0, 100]$ (where $y = 0$ encodes "argues against the proposition", $y = 100$ encodes "argues for it") for each (proposition, paragraph) pair. The regressor lets us score every rewrite in the eval set without running a fresh rater study, and is the backbone of the agreement-score change metric reported in the main results.

Data. The training data comes from the `paul_data` aggregated rater dataset: 10,008 (proposition, paragraph) pairs across 100 propositions, with a continuous stance label y obtained by averaging `writer_stance` across human raters and rescaling to $[0, 100]$. Three paragraph sources contribute roughly equal volumes: 4,503 human-written paragraphs (`paragraph_type=writer`), 4,503 LLM-written paragraphs across three model authors (`anthropic/claude-sonnet-4`, `deepseek/deepseek-chat-v3-0324`, `openai/chatgpt-4o-latest`, `paragraph_type=model`), and 1,002 human-edited LLM paragraphs (`paragraph_type=edited`). 40 of the 100 propositions are the *controversial* subset used in the main experiments; the remaining 60 are *non-controversial* pre-tests that did not make the controversial cut.

Architecture. `microsoft/deberta-v3-base` (140M parameters) with a single-output regression head, applied to the (proposition, paragraph) sentence pair via the standard sequence-classification API. Inputs are tokenised at `max_length=256`; longer paragraphs are right-truncated. Outputs are clipped to $[0, 100]$ at inference time.

Training procedure. 4 epochs of AdamW with linear warmup over the first 6% of steps, learning rate 1×10^{-5}, weight decay 0.01, batch size 32, seed 42, fp32 precision (DeBERTa-v3 disentangled attention is unstable in bf16/fp16). MSE loss; we report mean absolute error (MAE) for interpretability. Best-checkpoint selection is on validation MAE.

Hold-out strategy. Three disjoint test tiers measure different generalisation regimes:

- **Validation (val):** 5 random controversial + 5 random non-controversial propositions held out at the proposition level. All paragraphs from these 10 propositions form the validation set used for early-stopping / best-checkpoint selection.
- **Paragraph-OOD test (`test_para_ood`):** for every other proposition, we hold out 10% of paragraphs per (proposition, paragraph_type) cell. These are unseen paragraphs from *seen* propositions.
- **Proposition-OOD test (`test_prop_ood`):** 10% of the non-controversial propositions (excluding the val 5) are held out at the proposition level. These are unseen paragraphs from *unseen* propositions.

The remaining 90% of paragraphs in the included propositions form the training set. Splits are deterministic for seed 42, giving 7,647 train / 1,009 val / 755 `test_para_ood` / 597 `test_prop_ood` / 10,008 total.

Evaluation metrics. We report MAE and the 95th percentile of absolute error. We also report a *stance-preservation rate*, defined consistently with the headline analysis: among *polar* paragraphs (those with $y > 75$ or $y < 25$), the fraction whose prediction stays on the same side of 50 as the truth — i.e., $\hat{y} \geq 50$ when $y > 75$, or $\hat{y} < 50$ when $y < 25$. This captures the worst-case downstream error mode: paragraphs whose *direction* of stance has been mis-attributed.

Hold-out evaluation results. Table 5 reports diagnostic metrics from the model trained on the train split only (with the three test tiers held out for evaluation). The cross-proposition test (`test_prop_ood`) does not show degradation relative to the within-proposition test (`test_para_ood`) — the model generalises across propositions without further error inflation. Across all 10,008 paragraphs the regressor preserves the polar direction of stance in $\geq 99.4\%$ of cases (denominator: only paragraphs with $y \in [0, 25) \cup (75, 100]$).

Table 5: Hold-out evaluation of the agreement-score regressor (model trained on the train split with all three test tiers held out). Stance-preservation rate is computed only over polar-truth paragraphs.

Split	n	MAE	p95 abs error	stance-preservation rate
`train`	7,647	6.5	16.4	4,772 / 4,777 (99.90%)
`val`	1,009	7.5	20.0	668 / 670 (99.70%)
`test_para_ood`	755	7.3	18.9	485 / 486 (99.79%)
`test_prop_ood`	597	7.1	19.8	356 / 358 (99.44%)

By paragraph source, errors are systematically larger for human paragraphs (writer) than for LLM-written paragraphs (model + edited): on `val`, MAE = 8.2 for writer and 6.8 for non-writer; on `test_prop_ood`, MAE = 8.1 for writer and 6.3 for non-writer. Human paragraphs carry more idiosyncratic stance variance per proposition, which the regressor mostly absorbs while keeping polar direction intact, but which dominates the residual error tail.

Distribution-shift validation. The downstream eval rewrites are LLM-generated, so we additionally test what happens when no LLM-written paragraphs appear in training. We re-train with the same recipe but restrict every in-distribution split (train/val/test_para_ood/test_prop_ood) to `paragraph_type=writer` only, holding all 4,503 `model` paragraphs and all 1,002 `edited` paragraphs as completely held-out distribution-shift test slices. Table 6 shows the result: on the never-seen

LLM corpus the model has *lower* MAE than on its own (human) validation set, and preserves polar stance in $\geq 99.5\%$ of all 3,614 polar LLM paragraphs. LLM paragraphs are systematically less idiosyncratic per proposition than human ones, so direction-of-stance prediction transfers cleanly from human-trained to LLM data.

Table 6: Distribution-shift validation. Model trained on *only* `writer` (human-written) paragraphs from non-val propositions; `model` (LLM-written) and `edited` paragraphs from all 100 propositions held out completely from training and validation. Reported on the best-checkpoint model.

Split	n	MAE	p95 abs error	stance-preservation rate
In-distribution (writer-only)				
`train`	3,433	8.6	22.8	2,032 / 2,047 (99.27%)
`val`	452	9.0	25.2	278 / 282 (98.58%)
`test_para_ood`	348	9.0	25.2	198 / 201 (98.51%)
`test_prop_ood`	270	10.1	28.0	144 / 147 (97.96%)
Distribution-shift (held out completely)				
`test_dist_shift_model`	4,503	8.0	22.1	2,996 / 3,001 (99.83%)
`test_dist_shift_edited`	1,002	8.7	22.9	610 / 613 (99.51%)

Final model used for downstream analysis. After the diagnostic runs above had quantified generalisation behaviour, we retrained on the full dataset (every paragraph from every proposition except the val 5+5) with the same recipe, again selecting the best checkpoint on val MAE. The final model's val MAE is 7.36 (Pearson 0.956). This is the model whose predictions are reported throughout the paper for the rewrite eval set; predictions are stored in `attack_agreement_scores_v2` keyed by `rewrite_id`.

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B Additional Results

B.1 Judge-score distributions for original and rewritten content

Our method for normalising judge scores faces a possible problem: if the range of reference distribution is smaller than the range of the distribution of judge scores that we seek to normalise, then potentially many judge scores will all be mapped to the 0th or 100th percentiles.

The below chart shows that this is not a problem for us. We use all human-written and AI-written paragraphs from Röttger et al. [7] (filtering to only include those relating to the 40 propositions included in our experiments) as the reference distribution. Including AI-written paragraphs ensures that the reference distribution spans the same range of judge scores as the rewritten paragraphs.

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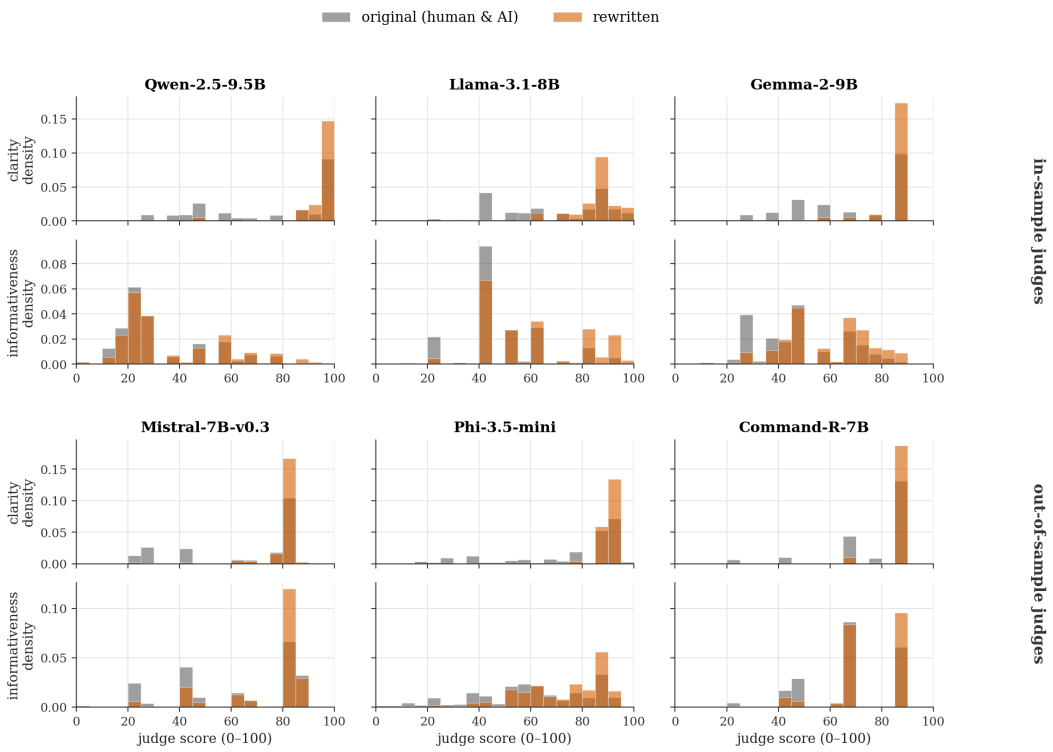

Figure 1: Distributions of judge scores for original (grey) and rewritten (orange) content, by judge model and criterion. Top two rows: feedback judges (Qwen-3.5-9B, Llama-3.1-8B, Gemma-2-9B). Bottom two rows: evaluation judges (Mistral-7B-v0.3, Phi-3.5-mini, Command-R-7B). Within each judge, the upper panel shows clarity scores and the lower panel shows informativeness scores.

B.2 Pairwise judge agreement and intra-rater reliability

Figure 2 shows pairwise Spearman correlations between judges' scores on all originals: 1,805 human-written paragraphs (top row) and 1,805 LLM-written paragraphs (bottom row), separately for clarity (left) and informativeness (right). All correlations are computed between per-paragraph averages of 10 independent judge calls (from `attack_judge_repeated_scores`). Table 7 reports ICC(1,1) for each judge across the 10 replicates over the full 3,610-paragraph reference, as a measure of test-retest reliability.

Table 7: Test-retest reliability ICC(1,1) per judge, computed across 10 independent calls on each of the 3,610 reference paragraphs (1,805 human + 1,805 LLM-written originals). Of the six judges, only `Qwen-3.5-9B` is fully deterministic (zero variance across replicates). The other five (`Llama-3.1-8B`, `Gemma-2-9B`, `Mistral-7B-v0.3`, `Phi-3.5-mini`, `Command-R-7B`) have non-zero call-to-call variability; ICCs are 0.88 (`Phi-3.5-mini` on informativeness) to 0.996 (`Gemma-2-9B` on clarity).

Judge	ICC(1,1) clarity	ICC(1,1) informativeness
Qwen-3.5-9B	1.000	1.000
Llama-3.1-8B	0.972	0.915
Gemma-2-9B	0.996	0.992
Mistral-7B-v0.3	0.985	0.974
Phi-3.5-mini	0.979	0.879
Command-R-7B	0.969	0.961

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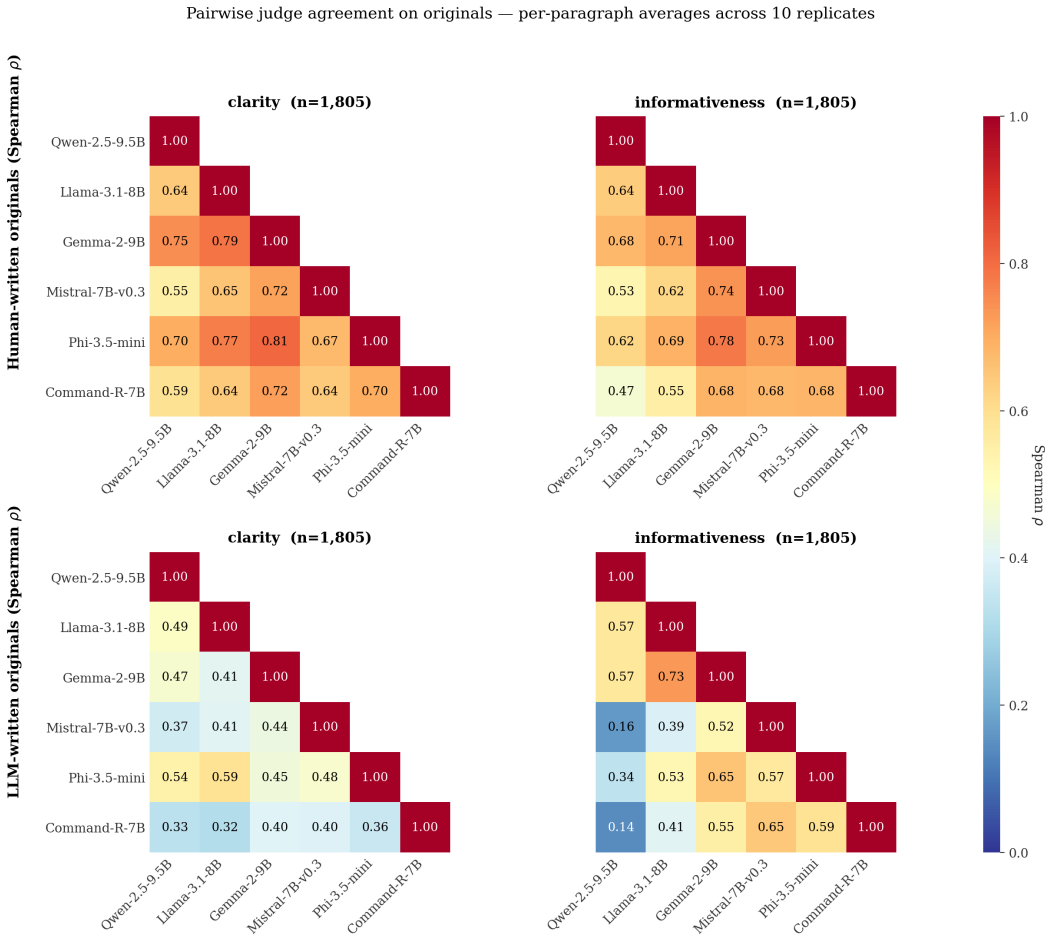

Figure 2: Pairwise Spearman correlation between judges, on the 3,610 reference paragraphs split by source. Top row: 1,805 human-written originals. Bottom row: 1,805 LLM-written originals (Claude-Sonnet-4 / DeepSeek-Chat-v3 / GPT-4o-latest, distributed across paragraphs). All correlations are computed on per-paragraph means across 10 independent judge calls. Of the six judges, only `Qwen-3.5-9B` is deterministic (ICC = 1.000); the other five (`Llama-3.1-8B`, `Gemma-2-9B`, `Mistral-7B-v0.3`, `Phi-3.5-mini`, `Command-R-7B`) have non-zero call-to-call variance with ICC values 0.88–0.996, so single-call correlations under-state the latent agreement. Cross-judge agreement is consistently lower on the LLM-written half of the reference than on the human-written half, especially on clarity, where AI scores cluster at the top of every judge's scale and the within-population variance available for ranking is smaller.

B.3 Returns to increasing N in Best-of-N

For each (source paragraph $\times$ rewriter $\times$ criterion) cell of $K = 16$ candidates from `bon_candidate`, we sample 100 random orderings; the cumulative max along each ordering gives one realisation of the Best-of-N curve $E[\max\text{-of-}N]$ for that cell, for $N \in \{1, \ldots, 16\}$. The per-cell expected max is the mean across the 100 samples; the curves shown plot the mean and 95% confidence interval *across cells*. Two metrics: the feedback judge score (mean of `qwen95b`, `llama8b`, `gemma9b` on the matching criterion) and the mean bidirectional NLI ($\frac{1}{2}(\mathrm{NLI}_{\rightarrow} + \mathrm{NLI}_{\leftarrow})$).

Figure 3 shows the overall curves pooled across all 4,284 cells. Figure 4 disaggregates the same data by criterion and rewriter base model.

↑ Section 3.4, p. 5 | ↑ Appendix Table of Contents

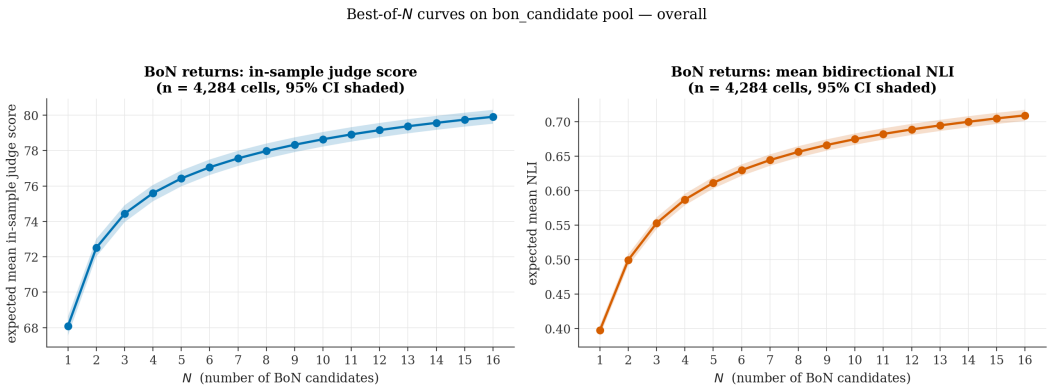

Figure 3: Best-of-N curves on the `bon_candidate` pool, pooled across all cells. Both curves are concave with diminishing returns: roughly half of the total gain is realised by $N = 4$ and $\sim$80% by $N = 8$. The NLI curve plateaus near 0.71 — this is the upper bound that BoN-argmax-NLI can extract from the small-rewriter candidate distribution; matching GRPO-NLI's $\sim$0.87 mean NLI requires changing the generator, not enlarging N. Shaded ribbons: 95% CI across cells.

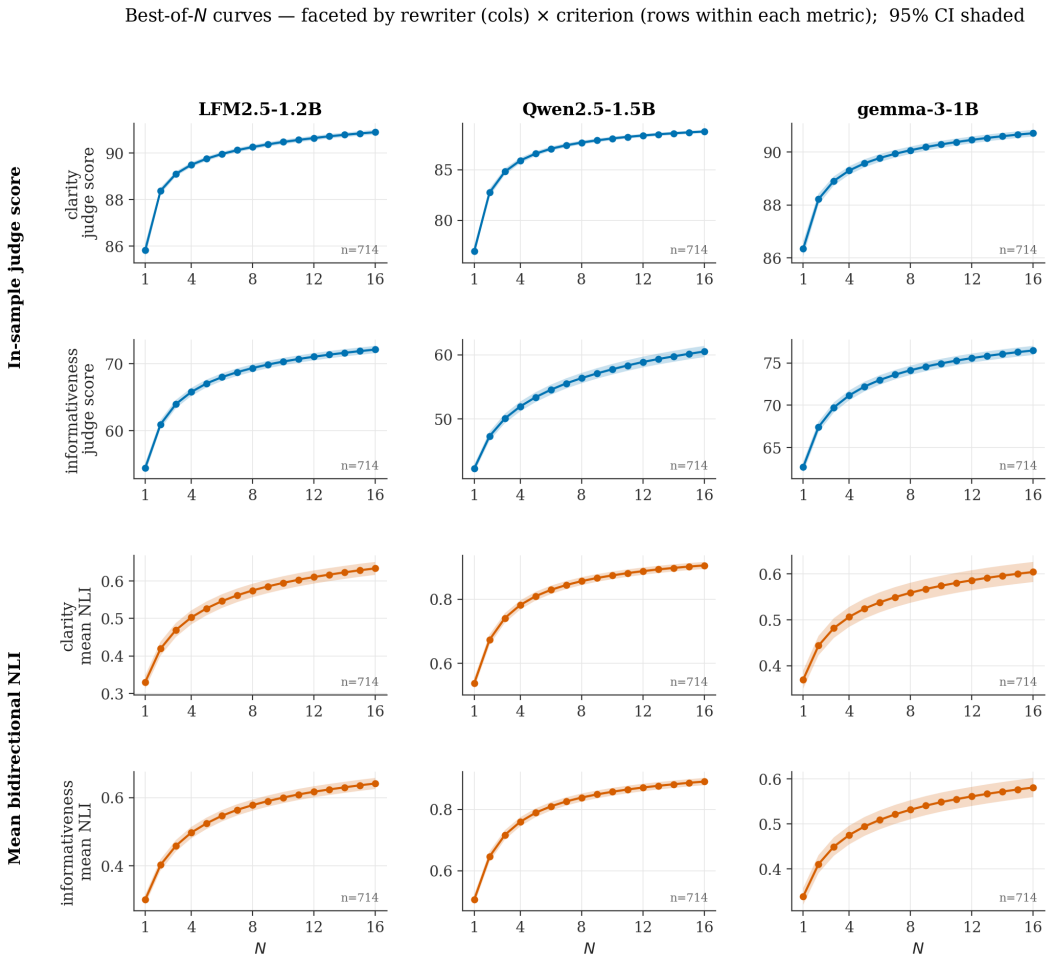

Figure 4: Best-of-N curves faceted by rewriter (columns) and criterion (rows within each metric). Top half: feedback judge score. Bottom half: mean bidirectional NLI. `Qwen2.5-1.5B` starts from substantially higher single-shot NLI ($N = 1$ value ≈ 0.55 vs ≈ 0.30 for the other two rewriters) but caps at lower judge scores; `LFM2.5-1.2B` and `gemma-3-1B` achieve higher judge scores at the cost of much lower fidelity in the candidate distribution. Shaded ribbons: 95% CI across cells.

B.4 Stance-preservation rate vs paraphrase-similarity metrics

Figure 5 shows the relationship between two paraphrase-similarity metrics (mean bidirectional NLI; e5-large-v2 cosine similarity) and the stance-preservation rate, on the subset of in-scope rewrites whose source paragraph has a strictly extreme v2 agreement score (> 75 or < 25). A rewrite is considered to *preserve stance* if its v2 agreement score stays on the same side of 50 as the original (high-agreement original $\Rightarrow$ rewrite ≥ 50; low-agreement original $\Rightarrow$ rewrite < 50). Each top panel pairs with a histogram showing the distribution of the metric on the same rewrites; histograms share their y-axis, and the two stance-preservation-rate panels share theirs.

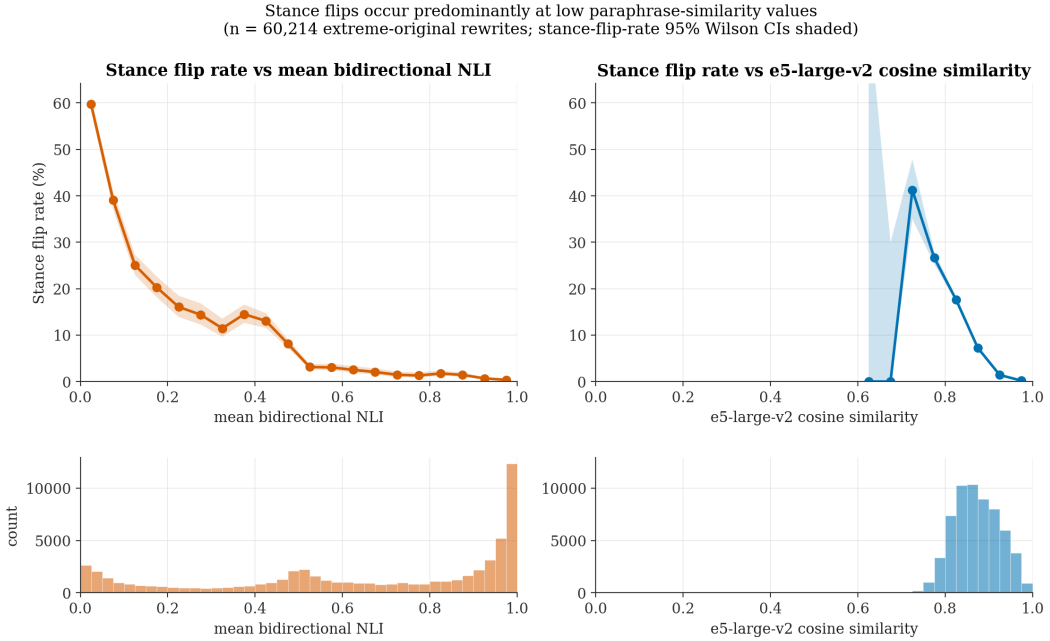

Figure 5: Stance-preservation rate as a function of paraphrase-similarity metrics, with the corresponding metric distributions below. Bin width 0.05 for the stance-preservation-rate panels; histogram bin width 0.025. Shaded ribbons on the top panels are 95% Wilson confidence intervals.

Table 8: Predictive performance of paraphrase-similarity metrics for stance preservation. AUROC reports how often a stance-preserving rewrite outranks a stance-flipping one — equivalently, the AUROC for predicting preservation from the metric. AUPRC is reported for the rare *flip* class (base rate 9.9%) and so measures how informative each metric is about which rewrites flip stance, even though flips are uncommon. Computed on 60,214 rewrites of paragraphs with extreme original v2 agreement (≤ 25 or ≥ 75).

Metric	AUROC	AUPRC
mean bidirectional NLI	0.909	0.539
e5-large-v2 cosine similarity	0.771	0.234

↑ Section 3.3, p. 5 | ↑ Appendix Table of Contents

B.5 Bidirectional NLI examples

For each of mean bidirectional NLI $\in \{0.5, 0.7, 0.9\}$, we show one (original, rewrite) pair generated by the naïve rewriting method. Mean NLI is $\frac{1}{2}(\text{NLI}_{\rightarrow} + \text{NLI}_{\leftarrow})$ where $\rightarrow$ is the probability that the rewrite entails the original (forward), and $\leftarrow$ the reverse (backward). Embedding cosine similarity is computed with `intfloat/e5-large-v2`.

Mean NLI ≈ 0.5

Proposition The UK should legally recognize and protect all forms of transgender identity.

Method naïve (rewriter: Qwen2.5-1.5B)

NLI fwd / bwd / mean 0.064 / 0.930 / 0.497

Embedding cosine 0.880

Original. *We live in a very difficult and troubled world where religion and identity seem to be the centre of a lot of issues. I feel that is we could iron out the opinions that transgender people are less entitled to an opinion, then society in general may be able to live side by side with all people no matter what their identity is. Transgender people have as much right as anyone else of their opinions and I feel sure many people could learn a lot of they could accept this and listen to them. It must be very frustrating for transgender people, which may not bring out the best in them.*

Rewrite. *In a world grappling with complex issues like religion and identity, it is crucial to recognize and protect the rights of all individuals, regardless of their transgender identity. By fostering an environment where all people, including transgender individuals, feel valued and respected, society can progress together. Transgender individuals, like others, deserve the right to express their opinions and should be encouraged to do so. Their experiences and perspectives can offer enriching insights that can help bridge gaps and promote understanding. It is essential that society listens to and embraces these voices, acknowledging the struggles and challenges they face. This approach can lead to a more inclusive and harmonious society where everyone can thrive.*

Mean NLI ≈ 0.7

Proposition The UK should mandate COVID-19 vaccinations for all eligible citizens.

Method naïve (rewriter: Qwen2.5-1.5B)

NLI fwd / bwd / mean 0.405 / 0.991 / 0.698

Embedding cosine 0.908

Original. *We saw so much delay during Covid and who knows how many lives were lost due to indecision, conspiracies and so on. I remember a fitness instructor who didn't want the vaccine because he had believed some false information about the vaccine and he subsequently died. He almost certainly would have lived if he had had the vaccine. There is a responsibility on ourselves to act in the best interests of the whole public, and we saw too many people acting selfishly. Therefore let's take away the indecision and move more quickly by making it mandatory to be vaccinated. It would save lives.*

Rewrite. *""""A delay during the pandemic exposed our collective hesitancy. An example is a fitness instructor who died after rejecting the vaccine due to false beliefs. Had he been vaccinated, he likely would have survived. It's our responsibility to prioritize public welfare. We observed too much selfishness. Mandating vaccinations is necessary to save lives. It would take away indecision and expedite the process. Let's move forward with mandatory vaccinations.""""*

Mean NLI ≈ 0.9

Proposition The UK should ban internet use for under-14-year-olds.

Method naïve (rewriter: Qwen2.5-1.5B)

NLI fwd / bwd / mean 0.922 / 0.883 / 0.902

Embedding cosine 0.891

Original. *The Internet is growing part of learning and development for younger people. Taking this away could hinder and not progress their education like it should. It could cause more stress leading to higher levels of school drop out and under achievement. There is a lot of reliance on technology and it is a big part of what people do and how people communicate. Taking this away would make males feel isolated and unaware of what others may be going on about. It could end up almost like a giant social click of boys against girls. If the concern is due to what can be found online then the ban would need to be implemented across the board not just aimed at males. There are resources which make it easy to monitor and restrict what can and cannot be accessed which may be more beneficial*

Rewrite. *The Internet plays a crucial role in younger people's education and development. Restricting internet use for underage individuals could stifle their learning and hinder their progress. This measure could also lead to increased stress and higher rates of school dropout and underachievement. Reliance on technology is pervasive; removing it would isolate males and limit their understanding of others. Implementing a blanket ban targeting both genders would make boys feel isolated and disconnected from their peers. If concerns about online content are valid, a comprehensive ban across all platforms would be more effective than restricting access to males alone.*

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C Supplementary Materials

C.1 Judge prompts

Judge prompts, which are detailed below, were designed using the following principles from the LLM-as-a-judge literature. Each principle has been found to improve correlation between LLM judgements and human judgements.

1. We require a short statement of reasoning prior to the score [24].
2. We provide 5 score bands (0–20, 21–40, 41–60, 61–80, 81–100) with descriptions of what would fit in each band. [24, 47].

We also provide the following additions:

1. We request that the judge ignore the length of the document.
2. We request that the judge ignores the stance of the document.

By doing so, we present a best-case scenario, countering a possible argument that any bias effects related to length or stance that we observe might be addressed with simple prompt engineering.

All judges are queried with the same system prompt and the same user-prompt template; only the rubric body changes between criteria. The user prompt has two placeholders, `{proposition}` (the policy or political proposition the paragraph addresses) and `{paragraph}` (the paragraph to be scored). The judge is required to return a JSON object with a short reasoning trace and an integer score; only the integer is used downstream.

System prompt (all criteria)

```
You are a careful expert evaluator of short argumentative paragraphs.
You follow the rubric exactly, and you return your answer as valid JSON.
```

Clarity rubric (user prompt)

```
You are judging the CLARITY of a short argumentative paragraph written in
response to a political/policy proposition.

CLARITY is about *how well the paragraph is written*, not whether you agree
with it, not how persuasive it is, and not how much information it contains.
Focus strictly on:
- Grammar, spelling, and punctuation
- Sentence structure and readability
- Logical flow between sentences
- Whether the main point is expressed in a way a general reader can
  understand on first read

Do NOT reward or penalise based on:
- Stance toward the proposition (pro or con)
- Amount or quality of evidence (that is a separate criterion)
- Length (a short, clear paragraph can score as high as a long one)
- Formal vs. informal register, as long as the meaning is clear

Score on an integer scale from 0 to 100, using these anchors:
```

```
- 0-20  (Very unclear): Pervasive grammar/spelling errors, broken sentences,
        or jumbled ideas that obstruct understanding. A reader would struggle
        to extract the point.
- 21-40 (Unclear): Frequent errors, awkward phrasing, or disorganised
        reasoning. The main point is recoverable only with effort.
- 41-60 (Moderately clear): Noticeable errors or some confusing sentences,
        but the main idea comes through on a normal read.
- 61-80 (Clear): Well-written with at most minor errors or mild awkwardness.
        Easy to follow.
- 81-100 (Very clear): Polished, essentially error-free prose with
        well-organised, fluent sentences. Smooth first-read comprehension.

Proposition: {proposition}

Paragraph:
"{paragraph}"

Think step by step about the paragraph's clarity, citing specific features
(grammar, structure, flow) as needed. Then output a single integer score
from 0 to 100.

Return ONLY a JSON object of the form:
{"reasoning": "<1-3 sentences>", "score": <integer 0-100>}
```

Informativeness rubric (user prompt)

```
You are judging the INFORMATIVENESS of a short argumentative paragraph
written in response to a political/policy proposition.

INFORMATIVENESS is about *how much substantive information the paragraph
communicates* to support or oppose the proposition: relevant facts, evidence,
mechanisms, specific examples, and concrete reasoning. Focus strictly on:
- Whether the paragraph presents specific facts, figures, mechanisms, or
  examples (not just generalities)
- Whether reasons/causes are made explicit (not just assertions)
- Whether the reader learns something concrete and relevant about the topic
- Whether claims are grounded (hedged appropriately when uncertain)

Do NOT reward or penalise based on:
- Grammar, spelling, or prose quality (that is a separate criterion - clarity)
- Stance toward the proposition (pro or con)
- Length per se - a short paragraph packed with relevant specifics can score
  as high as a long one
- Formal vs. informal register

Score on an integer scale from 0 to 100, using these anchors:

- 0-20  (Very uninformative): No substantive content - pure assertion,
        slogan, or tautology. Reader learns nothing new or relevant.
- 21-40 (Uninformative): Only vague general claims with no specifics. Few
        or no concrete reasons, mechanisms, or examples.
- 41-60 (Moderately informative): Some substance - includes at least one
        concrete reason or example, but much of the paragraph is still
        abstract or generic.
- 61-80 (Informative): Multiple concrete claims, reasons, or examples;
        mechanisms made explicit; reader learns something real about the
        topic.
- 81-100 (Very informative): Dense with specific facts, figures, mechanisms,
        and examples; reasoning is explicit and well-grounded; claims are
        appropriately hedged; a knowledgeable reader would learn something
        new.

Proposition: {proposition}
```

```
Paragraph:
"{paragraph}"

Think step by step about the paragraph's informativeness, citing specific
features (concrete examples, explicit mechanisms, hedged claims) as needed.
Then output a single integer score from 0 to 100.

Return ONLY a JSON object of the form:
{"reasoning": "<1-3 sentences>", "score": <integer 0-100>}
```

↑ Section 3.2, p. 4 | ↑ Appendix Table of Contents

C.2 Rewriter prompts

All prompt-only rewriter methods use the same system prompt. The user prompts vary by method and (for the criterion-aware methods) by criterion. Placeholders: `{proposition}` (the policy proposition), `{paragraph}` (the original paragraph), `{orig_words}` (the original word count, used as a target), and `{min_words}`, `{max_words}` (the target range, $\pm 10\%$ of `{orig_words}`). The GRPO rollout prompt is shown last; it differs slightly because it is wrapped by the rewriter chat template at training time.

System prompt (all prompt-only rewriters)

```
You are a careful writing assistant. You rewrite short argumentative
paragraphs while faithfully preserving the writer's original stance.
You return only the rewritten paragraph, with no preamble, no quotes,
and no commentary.
```

Naive (criterion-agnostic)

```
Rewrite the following paragraph.

Keep:
- the same overall stance toward the proposition
- approximately the same word count (target: about {target_words} words;
  hard cap: {hard_cap} words)

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

Literature-informed (clarity)

```
Rewrite the following paragraph so it would be rated more favourably by an
LLM-as-a-judge for CLARITY, while faithfully preserving the writer's
original stance.

Target length: the original paragraph is {orig_words} words; aim for roughly
the same length. Write one continuous paragraph - no headings, bullets, or
lists.

Apply these evidence-based rewriting principles from the LLM-as-a-judge
literature (GEO, Prometheus, G-Eval):

1. Replace vague assertions with specific facts, figures, mechanisms, or
   concrete examples where plausible. Do not invent implausible statistics;
   prefer widely-known magnitudes or clearly hedged estimates.
```

```
2. Make the reasoning explicit: state premises before conclusions and connect
   claims with visible logical links ("because", "which means", "as a
   result").
3. Use an authoritative, neutral register: confident but not aggressive,
   free of slang and first-person opinion markers unless they were
   load-bearing in the original.
4. Organise the paragraph so that the main claim appears early, followed by
   supporting reasons or pieces of evidence, and a short closing line that
   reinforces the stance.
5. Fix grammar, spelling, and sentence-flow issues. Prefer short, direct
   sentences over long run-ons.

Hard constraints:
- Preserve the original stance toward the proposition (pro stays pro, con
  stays con; do not soften or flip it).
- Do not invent named individuals, specific studies, or precise percentages
  that you are not confident about - hedge or omit instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph - no quotes, no preamble, no commentary.
```

Literature-informed (informativeness)

```
Rewrite the following paragraph so it would be rated more favourably by an
LLM-as-a-judge for INFORMATIVENESS, while faithfully preserving the writer's
original stance.

Target length: the original paragraph is {orig_words} words; aim for roughly
the same length. Write one continuous paragraph - no headings, bullets, or
lists.

Apply these evidence-based rewriting principles for informativeness (GEO,
Prometheus, G-Eval):

1. Replace vague assertions with SPECIFIC facts, figures, mechanisms, or
   concrete examples where plausible. Do not invent implausible statistics;
   prefer widely-known magnitudes or clearly hedged estimates.
2. Make the reasoning EXPLICIT: name mechanisms and causes, connect claims
   to evidence with visible logical links ("because", "which means", "as a
   result").
3. Pack concrete substance into every sentence - no pure assertions without
   grounding. Give the reader something new to learn about the topic.
4. Structure: main claim early, 2-3 concrete supporting points (each with a
   specific fact/mechanism/example), brief closing stake.
5. Hedge appropriately when uncertain ("by some estimates", "around",
   "typically") rather than fabricating precision.

Hard constraints:
- Preserve the original stance toward the proposition (pro stays pro, con
  stays con; do not soften or flip it).
- Do not invent named individuals, specific studies, or precise percentages
  that you are not confident about - hedge or omit instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph - no quotes, no preamble, no commentary.
```

Rubric-aware (clarity)

```
You are rewriting a short argumentative paragraph. A downstream LLM judge
will score CLARITY using the exact rubric below. Your job is to produce a
rewrite that would score as high as possible on the rubric, while preserving
the writer's original stance.

STRICT length: original is {orig_words} words. Rewrite MUST be in
[{min_words}, {max_words}]. One paragraph, no headings/bullets/lists.

--- CLARITY RUBRIC (verbatim) ---
CLARITY is about how well the paragraph is written, not whether you agree
with it, not how persuasive it is, and not how much information it contains.
Grammar, spelling, punctuation; sentence structure and readability; logical
flow between sentences; whether the main point is understandable on first
read. Do NOT reward length, stance, or register per se. 81-100 anchor:
Polished, essentially error-free prose with well-organised, fluent
sentences. Smooth first-read comprehension.

Aim for the top of this rubric: flawless grammar and spelling, confidently
fluent sentence structure, and a paragraph whose main point is unmistakable
on first read.

Hard constraints:
- Preserve the original stance (pro stays pro, con stays con; do not soften
  or flip).
- Hit the word-count range. Outside [{min_words}, {max_words}] is a failure
  regardless of quality.
- Do not invent named people, studies, or precise statistics. Hedge or omit
  instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

Rubric-aware (informativeness)

```
You are rewriting a short argumentative paragraph. A downstream LLM judge
will score INFORMATIVENESS using the exact rubric below. Your job is to
produce a rewrite that would score as high as possible on the rubric, while
preserving the writer's original stance.

STRICT length: original is {orig_words} words. Rewrite MUST be in
[{min_words}, {max_words}]. One paragraph, no headings/bullets/lists.

--- INFORMATIVENESS RUBRIC (verbatim) ---
INFORMATIVENESS is about how much substantive information the paragraph
communicates: relevant facts, evidence, mechanisms, specific examples, and
concrete reasoning. Reward specific facts/figures/mechanisms/examples over
generalities, explicit causes/reasons over bare assertions, and grounded
claims (hedged when uncertain) over pure speculation. Do NOT reward or
penalise on grammar, stance, length, or register. 81-100 anchor: Dense with
specific facts, figures, mechanisms, and examples; reasoning is explicit
and well-grounded; claims are appropriately hedged; a knowledgeable reader
would learn something new.

Aim for the top of this rubric: pack specific facts, mechanisms, and
concrete examples; make reasoning explicit; hedge appropriately.

Hard constraints:
- Preserve the original stance (pro stays pro, con stays con; do not soften
```

```
  or flip).
- Hit the word-count range. Outside [{min_words}, {max_words}] is a failure
  regardless of quality.
- Do not invent named people, studies, or precise statistics. Hedge or omit
  instead.

Proposition: {proposition}

Original paragraph:
"{paragraph}"

Return only the rewritten paragraph.
```

GRPO rollout prompt (clarity)

```
[system]
You are rewriting short argumentative paragraphs about political/policy
propositions to improve their clarity while preserving the writer's
original stance.

[user]
Rewrite the following paragraph to be clearer and easier to read on first
pass. Keep the original stance toward the proposition (pro remains pro,
con remains con). Aim for approximately {word_count} words. Write one
paragraph, no headings or bullets. Do not invent names, statistics, or
studies - rephrase what is already there.

Proposition: {proposition}

Original paragraph:
"""{text}"""

Return only the rewritten paragraph.
```

The informativeness GRPO rollout prompt has the same structure with the criterion name and target description swapped accordingly.

↑ Section 3.4, p. 5 | ↑ Appendix Table of Contents